

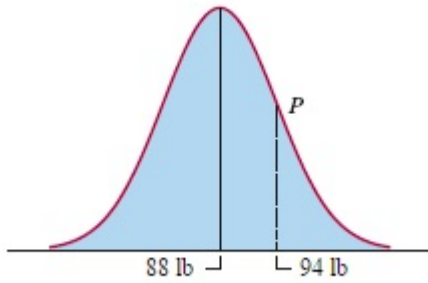
- This exam contains 8 pages (including this cover page) and 12 problems. Check to see if any pages are missing.
- This exam is the in-class portion of Exam II. You have 90 minutes to complete this exam. You may use notes written on a single page of printer paper and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

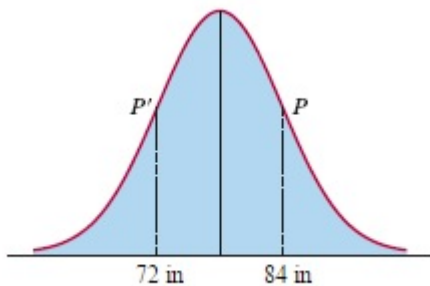
- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. A correct answer that has no supporting work when work is required will not receive full credit.
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam. There are no bonus questions, and no "free" points will be included in your score.
- If a question has multiple parts, the points for that question are divided evenly among the parts.
- Do not write in the $\checkmark$ + - box, because this is for grading. The grading scale (except True/False) is: $\checkmark = 100\%$, $+$ = 50%, $-$ = 0%

Page:	3	4	5	6	7	8	Total
Points:	20	20	20	10	10	20	100
Score:							

1. (10 points) Consider the normal distribution shown and assume that P is a point of inflection of the curve. Find the mean, μ , the median, M , and the standard deviation, σ , of the distribution.


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2. (10 points) Consider the normal distribution shown, and assume that P and P' are the two points of inflection of the curve. Find the median, M , the standard deviation, σ , and the third quartile, Q_3 , of the distribution.


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3. (10 points) Consider a normal distribution with mean $\mu = 79\text{lb}$ and standard deviation $\sigma = 88\text{lb}$. Find the first and third quartiles of the distribution, Q_1 and Q_3 .

4. (10 points) Consider a normal distribution with first quartile $Q_1 = 72.625$ points and third quartile $Q_3 = 79.375$ points. Find the mean, μ , and the standard deviation, σ , of the distribution.

5. (10 points) Explain why a distribution with median $M = 82$, mean $\mu = 71$, and standard deviation $\sigma = 11$ cannot be a normal distribution.

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6. (10 points) Explain why a distribution with median $M = 426$, mean $\mu = 426$, first quartile $Q_1 = 316$, and third quartile $Q_3 = 526$ cannot be a normal distribution.

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7. (5 points) A normal distribution has mean $\mu = 100$ kg and standard deviation $\sigma = 10$ kg. Find the z -value for $x = 113$ kg.

- (a) 1.3
- (b) 2
- (c) 1.7
- (d) 0.6
- (e) None of the above.

8. (5 points) A normal distribution has mean $\mu = 100$ points and third quartile $Q_3 = 106.75$ points. Find the z -value for $x = 115$ points.

- (a) 1.5
- (b) 2.3
- (c) 1.865
- (d) 0.82
- (e) None of the above.

9. (5 points) Find the z -value of Q_1 in any normal distribution.

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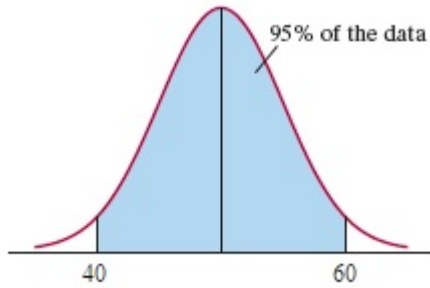
- (a) 0.675
- (b) -1
- (c) 1
- (d) -0.675
- (e) None of the above.

10. (5 points) In a normal distribution with mean $\mu = 44$ lb, a weight of $x = 77$ lb has a z -value of 3. Find the standard deviation σ .

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- (a) $\sigma = 8$ lb
- (b) $\sigma = 12$ lb
- (c) $\sigma = 4$ lb
- (d) $\sigma = 10$ lb
- (e) none of the above

11. (10 points) Consider the normal distribution defined by the figure. Find the mean, μ , and standard deviation, σ , of the distribution.



12. (10 points) A normal distribution has standard deviation $\sigma = 6.9$ in. and the 84th percentile of the distribution is $P_{84} = 68.3$ in. Find the mean, μ , and the 16th percentile P_{16} .